Arunabha Majumder

Mechanical Design Engineer

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Volvo Construction Equipment, Powertrain Technology, Eskilstuna

Application: Mechanical Design Engineer, Base Engine systems (Req. 36606)

Dear Hiring Team,

I am applying for the Mechanical Design Engineer role in the Base Engine and Auxiliaries team. I am a mechanical engineer doing my PhD in Mechatronics and Robotics at Aalborg University, with my thesis expected to be finished in March. The core of this job, creating 3D models, drawings and engineering reports and improving designs over time, is what I have been doing every week for the last 3 years.

My main project is a hybrid shoulder exoskeleton. The design idea was mine: a spring carries most of the arm load so a small 6 Nm motor is enough. I modelled it in SolidWorks, made the drawings, checked the parts with stiffness analysis in SolidWorks and optimised the design with a genetic algorithm. Then I 3D printed and rebuilt the parts many times, improving them after each test. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I wrote up the work in reports and papers and worked closely with colleagues on control, electronics and software.

I want to be clear about what I have not done yet. I have not worked with cast parts or engines, and my CAD experience is in SolidWorks, not Creo or CATIA. I learn new CAD tools quickly, and I would really like to learn design for casting from an experienced team. In my Bachelor's I did the CAD, machining and assembly of a rehab exoskeleton myself, so I know how a drawing turns into a real part.

I also have a Diploma in Fine Arts, so I often explain an idea with a quick sketch, which helps a lot when working with suppliers and other teams. The move towards non-fossil fuels in your engines is a big reason I would be glad to contribute here.

Thank you for considering my application. My portfolio is at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder